

Capitolis Counterparty Credit Risk Engine

Methods, Results, and How Every Round of Feedback Was Incorporated

Berkeley MFE Industry Project

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The book: 16 trades, 3 counterparties

- 8 equity total-return swaps
- 4 bond forwards, 4 bond total-return swaps
- 37 equities (US + Tokyo) + USDJPY

Every input is real, sourced data

- Equities/dividends: Yahoo Finance
- USD curve: bootstrapped from real Databento SOFR futures
- Treasury yields: FRED
- Long end + vol cube: licensed Bloomberg data, spliced in and cross-checked against public sources

Exposure definition (the brief):

$$\max(V(t+10bd) - V(t-1bd), 0)$$

- Signed prior-day variation margin
- No initial margin, no threshold
- Netted per counterparty only

Secondary view: uncollateralized level exposure $\max(V(t), 0)$ – a no-margin upper bound

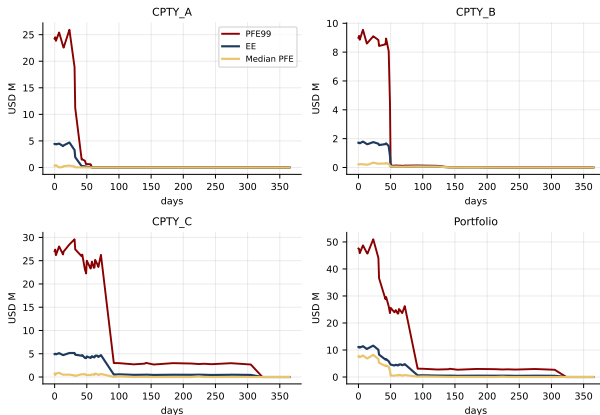
- **Equities & FX:** correlated geometric Brownian motion, real 39×39 correlation matrix (3-year realised)
- **USD rates:** one/two-factor Hull-White; mean reversion fitted to the swaption market, volatility fitted to realised *long-end* Treasury vol (0.96%, $a=0.0167$)
- **JPY rates:** a second, independent Hull-White factor from the real JPY OIS curve – so JPY names and USDJPY move with real JPY rate risk, not a constant assumption
- **Sampling:** Latin Hypercube throughout; every sensitivity reuses the identical random draws in the base and shocked run (common random numbers)

Scenarios (standard)	5,000
Reporting horizon	1 year

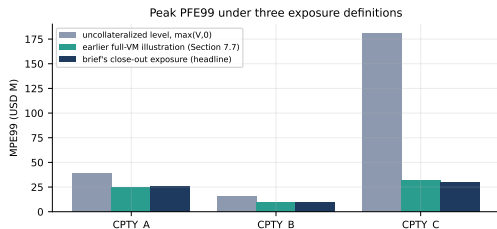
Results (5,000 scenarios, close-out exposure)

	MPE99	Peak EE
CPTY_A	\$25.9M	\$4.7M
CPTY_B	\$9.6M	\$1.8M
CPTY_C	\$29.6M	\$5.2M
Portfolio	\$51.0M	\$11.6M

- Portfolio MPE99 = 78% of the sum (\$65.1M): no netting across counterparties, but exposures peak in the same weeks and are positively correlated
- CVA \$11.7k close-out / \$247k level
- SA-CVA capital \$0.10M / \$2.23M
- SA-CCR EAD \$260M

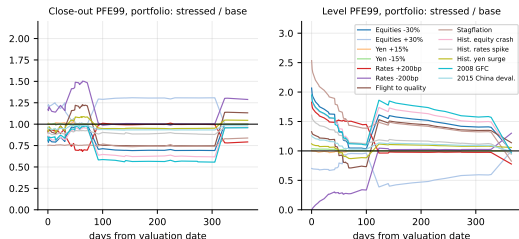


Close-out vs. level exposure: two brackets on the truth



- **Close-out** (the brief, primary): only the 10-day move from an already-margined start. Portfolio MPE99 **\$51.0M**.
- **Level** (secondary): whole mark-to-market at risk, no margin. Portfolio MPE **\$201.9M** – driven by CPTY_C's bond forward.
- The real answer sits between the two, depending on the actual CSA's threshold, MTA, and initial margin – none of which were considered.

Stress testing methodology



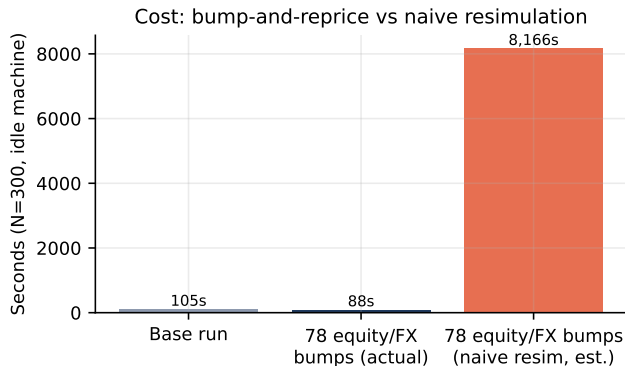
13 fixed scenarios, same random draws as the base case:

- 8 hypothetical: equities $\pm 30\%$, yen $\pm 15\%$, rates $\pm 200\text{bp}$, flight-to-quality, stagflation
- 5 historical replays, chosen by rule (worst 10-day window), not by hand

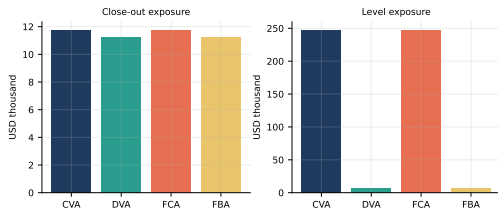
Read at every reporting date, every trade, every counterparty.

Sensitivities (Greeks): method

- Bump-and-reprice with **common random numbers**: same draws in base and bumped run, so the difference is signal, not noise
- Equity/FX bumps: rescale the simulated GBM paths exactly - **no resimulation**
- Rate/vol bumps: require a genuine resimulation
- Pathwise (closed-form) estimator implemented for equity/FX as a fast cross-check



CVA, DVA, FVA: the credit and funding adjustments



- **CVA:** expected loss from counterparty default, credit-triangle default probabilities from BBB bond-index spread proxies
- **DVA:** mirror image - our own default risk on what we owe them
- **FVA:** funding cost of positive exposure less funding benefit of negative exposure
- **SA-CVA:** Basel standardised-approach capital from CVA's own market-risk sensitivities

Feedback tracker: everything raised, everything addressed

Earlier meeting (5 items)

1. Robust stress testing **Addressed**
2. Sensitivities: method, cost, faster alternative **Addressed**
3. PCA explainability and speed **Addressed**
4. Final results explanation **Addressed**
5. CVA walkthrough and set-up **Addressed**

This meeting (6 items)

6. 2008 and more testing periods **Addressed**
7. LHS to calculate sensitivities **Addressed**
8. KVA addition **Addressed**
9. CVA is for sensitivities **Addressed**
10. DV01 via a Jacobian **Addressed**
11. Excel parametric vol-bump test **Addressed**

The next 11 slides take each item in turn: what was asked, what we did, the results, and the conclusion.

1. Robust stress testing

Feedback: make it a robust, structured test, not a paragraph – data, inputs, sensitivity along the path, all products

What we did

- 13 scenarios (8 round-number, 5 historical), each a full re-run of the Monte Carlo from the shocked state, same random draws as base
- Results read at every reporting date, every trade, every counterparty
- A per-trade heat map shows exactly which products move under which shocks

Result / finding

- Close-out MPE99 moves at most -24% to $+23\%$; level MPE99 moves up to $+89\%$ - margin is why
- Equity swaps respond only to equity/FX shocks; bond forwards/TRS respond only to rate shocks - confirms trade structure, not a coincidence

Conclusion: stress sensitivity is now quantified, dated, and attributable to a specific shock.

2. Sensitivities: method, cost, faster alternative

Feedback: explain how Greeks are generated, what they cost, and whether something faster exists

Method

- Bump-and-reprice, common random numbers (CRN)
- CRN cuts estimator noise from \$245k to \$1k – 282x

Cost (N=300, 8 cores)

- Base run: 105s
- 78 equity/FX bumps: 88s (vs. ~8,200s naive – 93x cheaper)
- Rate/vol bumps: still need resimulation, ~109s each

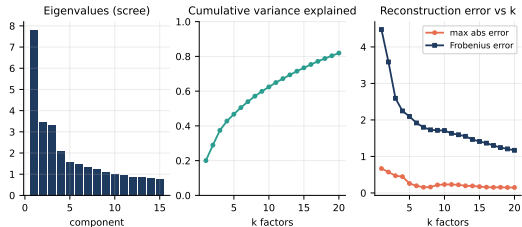
Faster alternative

- Pathwise (closed-form) delta for equity/FX: 0.2–0.4s vs. ~260s for bump-and-reprice
- Validated against bump-and-reprice: median 0.02%, worst 0.4% difference

Conclusion: common random numbers is what actually controls Greeks noise; pathwise is a validated, production-ready shortcut for linear (equity/FX) risk only.

3. PCA explainability and speed

Feedback: improve the explainability of the PCA factor model, and quantify the speed it adds



Explainability

- Factor 1 (20% of variance): a broad market factor
- Factor 2 (8.9%): Japan vs. US
- 5 factors explain 47%; 10 factors explain 62%

Speed: measured honestly, found to be none

- Full matrix: 1.91s per correlated-shock step
- 5-factor PCA: 2.08s; 10-factor: 2.29s – *slower*, not faster

Conclusion: kept the full matrix as default; PCA stays as an explainable robustness check, not a speed optimisation.

4. Final results: what drives each counterparty

Feedback: confirm A=equity, C=rates, bonds vs. equities inversely, and why the portfolio is so close to additive

	Equity $\Delta/+1\%$	DV01/ $+1bp$
CPTY_A	-\$2.25M	\$31k
CPTY_B	-\$0.96M	-\$4k
CPTY_C	-\$1.24M	\$594k

Rank correlation of exposures: A-B 0.34, A-C 0.33, B-C 0.12

- A is the most equity-driven; C is the rate netting set (one \$500M short forward on a 2049 Treasury) – but C also carries meaningful equity delta
- Yield-equity correlation in our data is ≈ 0.00 – the “inverse” relationship is regime-dependent, not fixed
- Portfolio MPE99 (\$51.0M) is close to A+C (\$54.4M): nothing nets *across* counterparties by construction, and the three books peak in the same weeks

5. CVA walkthrough and set-up

Feedback: walk through the CVA calculation in full, and set up the open questions for next time

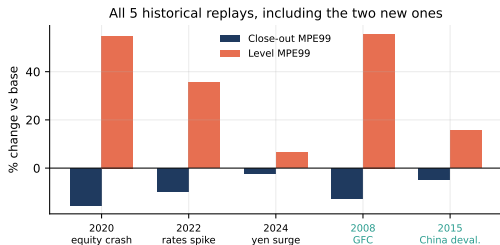
Steps (CPTY_C example)

1. Exposure profile from the simulation, discounted along each path
2. Default probability from a credit spread (BBB proxy: 59bp at 1y, 91bp at 5y) via the credit triangle
3. LGD 60%; multiply and sum

Results

- CPTY_C close-out CVA \$8k; total \$12k
- Rating sensitivity: AA \$7k, A \$8k, BBB \$12k, BB \$18k

6. Add more testing periods



Method: extended the public price history back to 2007 (28 of 37 names covered) and found the worst 10-day equity window inside two named crises, using the same objective rule as the existing replays.

- **2008 GFC:** Sep 29 – Oct 10, 2008; –23% median / –44% worst single-name return
- **2015 China deval.:** Aug 2015, –6.8% median

Result: 2008 moves close-out MPE99 by –13%, level by +56% – inside the range the existing scenarios already covered. **Conclusion:** leaving out 2008 was not quietly understating the tail.

7. Use Latin Hypercube to calculate a sensitivity

Method

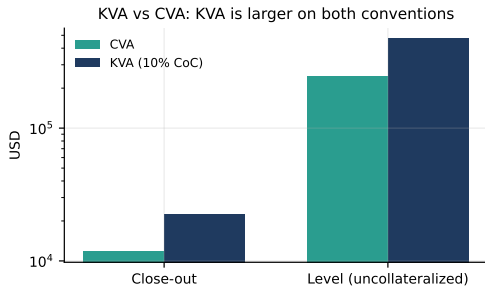
- In one-factor Hull-White, $r(t) = x(t) + \alpha(t)$
- $x(t)$: the simulated stochastic factor – depends only on the random draws, a, σ
- $\alpha(t)$: purely deterministic, depends only on the curve
- A curve bump (fixed a, σ) leaves $x(t)$ **identical** to the base run

Result

- The whole bump effect is one closed-form number per date, added to the base run's own paths
- **No resimulation. No new random draws.**
- Validated against a true independent resimulation: agreement to 10^{-15} relative precision

Conclusion: yes - exactly, not approximately, for any curve-only bump under the one-factor model.

8. KVA addition



Method: added KVA as a fourth, explicitly illustrative xVA term:

$$K(t) = 8\% \times RW \times \alpha \times EPE(t)$$

discounted and integrated like CVA/DVA/FVA, scaled by an assumed cost of capital (8/10/12% swept – none given).

Result at 10% CoC:

- KVA \$22.4k (close-out) vs. CVA \$11.7k
- KVA \$470k (level) vs. CVA \$247k

Conclusion: KVA exceeds CVA because it scales the whole EPE profile by a fixed ratio, not a small IG default probability – explicitly illustrative, pending a real capital methodology.

9. CVA for understanding sensitivities

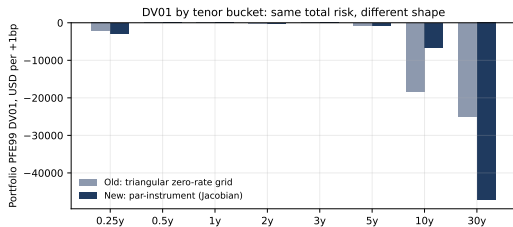
What this means in practice

- The headline CVA number is small and not the point
- What a credit/XVA desk actually manages day to day is the *sensitivity* of CVA to market and credit inputs

What we deliver as a result

- CVA01: sensitivity to a 1bp move in the counterparty spread
- Full rating table: AAA \$5.0k → B \$32.8k
- SA-CVA weighted sensitivities by risk class (rates, FX, equity, credit spread)

10. DV01 via a Jacobian



Old method: bump the *fitted* zero curve at 8 hand-picked tenors, triangular in between.

New method: bump the curve's own ~45 native construction pillars (SOFR futures + Bloomberg long end), grouped into the same 8 buckets – the genuine par-instrument sensitivity.

Result: the 30y bucket carries **82%** of portfolio DV01 (new) vs. **54%** (old) – the old grid smeared the 20–50y risk (where the 2049 bond forward lives) into the 10y bucket. Bucket sums match the parallel bump to within 0.1% either way.

11. Parametric test in Excel: bump equity vol

Method

- Bump one USD equity's vol $\pm 1\%$
- Compute a simple ATM position's expected exposure $E[\max(S_T - S_0, 0)]$ two ways: closed-form Black formula (live Excel formula), and Monte Carlo

Results

- Level match: within $\sim 3\%$
- Vega match: within $\sim 7\%$
- (Realised vol formulas match calibration to 0.1%; simulated vol matches analytic to 1–2%)

The kickoff deck flagged two extra-credit opportunities beyond the core engine: “*Extra Credit Opportunity: xVA*” under Exposure and Greeks, and “*Risk bonds + CDS data for new sample trades*”.

xVA outputs

- CVA, DVA, FVA and **KVA** – all four, on both exposure conventions
- Basel SA-CVA capital and SA-CCR exposure at default

	Close-out	Level
CVA	\$11.7k	\$247k
DVA	\$11.2k	\$7.5k
FVA	\$0.5k	\$240.0k
KVA (10% CoC)	\$22.4k	\$470k

SA-CVA capital \$0.10M / \$2.23M; SA-CCR EAD \$260M.

Risky bonds + CDS for a new sample trade

- A sample risky-bond forward (25M notional, 5% BBB corporate; not part of the ESF book), priced and simulated with a real issuer credit curve, re-anchored at every node
- Issuer curve: ICE BofA BBB option-adjusted spreads, credit-triangle survival, deterministic spreads (same construction as the counterparty CVA proxy)

Limitation, disclosed: no real CDS quotes were obtainable, so the issuer curve is a bond-index proxy, not actual CDS – delivered in spirit, not the full ask.

Thank you

Questions and discussion